Investment in Storage and Hydrogen

Group 15 | Technical report, Tier 1 | 2026/27

Executive summary

Under economic dispatch in 2023, one 20 MW electrolyser and no battery add £8.57 million per year in equivalent value at a hydrogen netback of £7.799/kg and a daily offtake limit of 10 t. If electricity sales have daily priority, the preferred investment is zero. Of the 171.84 GWh used by the electrolyser under economic dispatch, 13.92 GWh reduces unused generation and 157.92 GWh displaces direct sales. Investment depends on the buyer's terms and permission to redirect saleable electricity.

1 Decision context and data

The constructed system combines the supplied portfolio and simulated wind farm [1]. Generation is fixed, demand caps sales, and neither imports nor revenue from unused electricity enter the model.

We align five series to 2,452 UTC days (1 January 2020–17 September 2026) and reconcile fuel totals. Repairs replace midnight snapshots with full-day records, remove zero-volume price entries and flag estimated gaps [2]. Daily energy is mean MW multiplied by 24 h. Prices are in 2020 pounds using monthly consumer price indices [3]. We size assets on 2023.

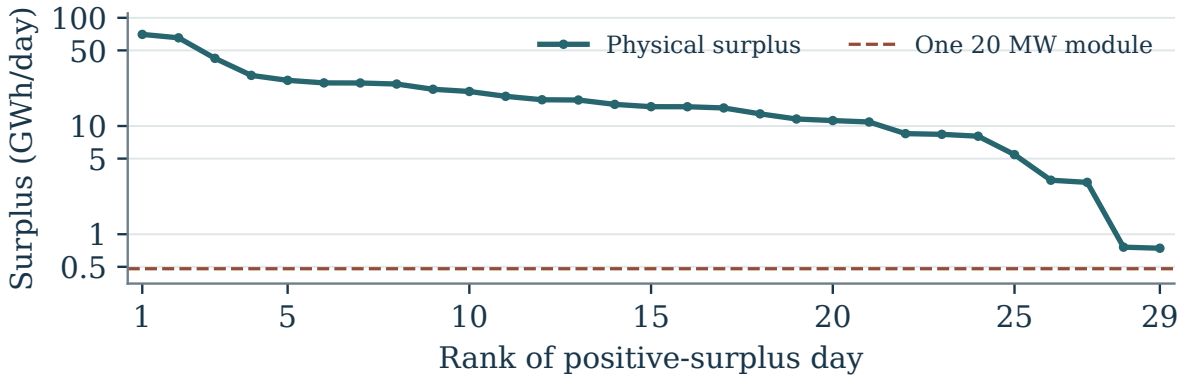


Figure 1. Physical surplus in 2023. The 29 days with positive surplus total 549.94 GWh. A 20 MW module can use at most 0.48 GWh per day.

2 Tier 1. Flexibility investment

2.1 Technology and economic assumptions

The electrolyser capital cost uses the International Energy Agency midpoint. Fixed operating and stack costs use UK reference estimates [4, 5, 6]. The netback after non-power variable costs uses a rebased policy benchmark [7, 3]. The offtake ceiling is a scenario.

Table 1. Reference technology and valuation assumptions in 2020 pounds.

Input	Assumption
Battery	1.5 MW / 3 MWh per module. 20–80% state of charge, 90% round-trip efficiency, £256,235/MWh installed, £14,862/MW-year fixed cost, 15-year life.
Electrolyser	20 MW per module, 20–100% online load, 55.9 kWh/kg, £29.21m installed and £0.517m annual fixed cost per module. 30-year life with £9.79m stack renewals in years 11 and 22.
Finance	5% real discount rate, £7.799/kg netback and 10 t/day offtake ceiling.

2.2 Mixed-integer formulation

Let $t \in \mathcal{T}$ index the days. The inputs G_t, D_t are supply and demand in MWh, p_t is the wholesale value in £/MWh, v is the hydrogen netback in £/kg and L the daily offtake limit in kg. The decision variables q_t, c_t, r_t, h_t, u_t are direct sales, charging, discharge, electrolyser input and unused generation in MWh. The installed battery and electrolyser counts are b, n, and z_t is the number of online electrolyser modules. The energy requirement is $\kappa = 0.0559$ MWh/kg. Let Y be the number of years represented by the sample (one for a complete calendar year), and a_B, a_H the equivalent annual costs per module.

$$\max \quad \frac{1}{Y}\sum_{t\in\mathcal{T}}\left[p_t(q_t + r_t) + v\frac{h_t}{\kappa}\right] - a_B b - a_H n, \tag{1a}$$

$$\text{subject to} \quad q_t + c_t + h_t + u_t = G_t, \qquad q_t + r_t \le D_t, \tag{1b}$$

$$c_t \le \frac{E_B}{\eta} b, \qquad r_t \le \eta E_B b, \qquad \sum_{t\in\mathcal{T}} r_t = \eta^2 \sum_{t\in\mathcal{T}} c_t, \tag{1c}$$

$$\underline{E}_H z_t \le h_t \le \overline{E}_H z_t, \qquad h_t/\kappa \le L, \qquad z_t \le n, \tag{1d}$$

$$q_t, c_t, r_t, h_t, u_t \ge 0, \qquad b, n, z_t \in \mathbb{Z}_{\ge 0}. \tag{1e}$$

The day-indexed constraints hold for all $t \in \mathcal{T}$. Equation (1b) allocates generation and limits electricity sales. The usable storage window is $E_B = 1.8$ MWh per module and each charge or discharge leg has efficiency $\eta = \sqrt{0.90}$. Equation (1c) permits one usable window of throughput per day and balances energy over the year. In (1d), $\underline{E}_H = 96$ and $\overline{E}_H = 480$ MWh are the daily loading limits per online module. Policy A uses the model above. Policy B also imposes $q_t = \min\{G_t, D_t\}$ each day.

For either technology $j \in \{B, H\}$, the equivalent annual cost is

$$a_j = F_j + \frac{i}{1-(1+i)^{-N_j}}\left(K_j + \sum_{k\in\mathcal{R}_j}\frac{C_{jk}}{(1+i)^k}\right), \tag{2}$$

where i is the real discount rate, N_j the service life, F_j annual fixed cost, K_j upfront investment including any commissioning charge, and C_{jk} the renewal cost in year k of the renewal set $\mathcal{R}_j$. Each reported gain subtracts the objective value of the zero-investment case under the same policy.

2.3 Portfolio results and validation

FICO Xpress solves the mixed-integer program. Table 2 shows the 2023 decisions. With one fixed 20 MW electrolyser, a simple sell-first rule uses only the remaining electricity when it can meet minimum loading. Its annual gain is −£1.07m. Economic dispatch improves this result by £9.64m per year at the same installed capacity.

Table 2. 2023 installed capacity and equivalent annual gain relative to the policy-matched zero-investment case. Capacity is shown as battery / electrolyser in MW.

	A. Economic dispatch		B. Daily electricity priority	
Scenario	Capacity	Gain (£m/yr)	Capacity	Gain (£m/yr)
No investment	0 / 0	0.00	0 / 0	0.00
Battery only	0 / 0	0.00	0 / 0	0.00
Electrolyser only	0 / 20	8.57	0 / 0	0.00
Joint capacity choice	0 / 20	8.57	0 / 0	0.00
Fixed 20 MW module	0 / 20	8.57	0 / 20	-1.07

Battery economics. Even with free charging, selling a full usable discharge on every positive-price day in 2023 yields at most £48.7k per module, below the £96.4k annual cost. This bounds the value of adding a battery under either policy without a minimum annual sales requirement.

Chronological check. Forcing one module in A yields £13.3k in operating value in the annual pool. To enforce chronological operation, we impose

$$s_t = s_{t-1} + \eta c_t - r_t/\eta, \qquad 0 \le s_t \le E_B b, \qquad s_0 = s_{|\mathcal{T}|} = 0, \tag{3}$$

where s_t is usable inventory after day t. Operating value falls to £4.3k, below even the lower fixed-cost case of £11.1k/year. Intraday and ancillary-service revenue sit outside this daily valuation.

Energy attribution. Of the 171.84 GWh used by the 20 MW plant, 13.92 GWh reduces unused generation and 157.92 GWh displaces direct sales. The 13.92 GWh is 8.1% of the electrolyser input and 2.53% of physical surplus. Under a surplus-first allocation, $C_t = \min\{h_t, \max(G_t - D_t, 0)\}$. With wind and solar generation R_t, their contribution W_t to captured surplus satisfies

$$\max\{0, C_t - (G_t - R_t)\} \le W_t \le \min\{C_t, R_t\}. \tag{4}$$

Summing the daily bounds gives 0–13.92 GWh. Including biomass and run-of-river hydro leaves this interval unchanged. A narrower estimate requires evidence on how generation is allocated.

2.4 Contract terms and service requirements

Figure 2 varies the netback and daily acceptance cap together. At 10 t/day, the 20 MW module breaks even at £4.61/kg in A and £12.08/kg in B. In A, installing 40 MW under the same cap increases hydrogen sales but lowers the gain to £7.47m/year. At 50 t/day, A chooses 120 MW and gains £49.09m/year on £175.25m of initial capital, conditional on the larger contract.

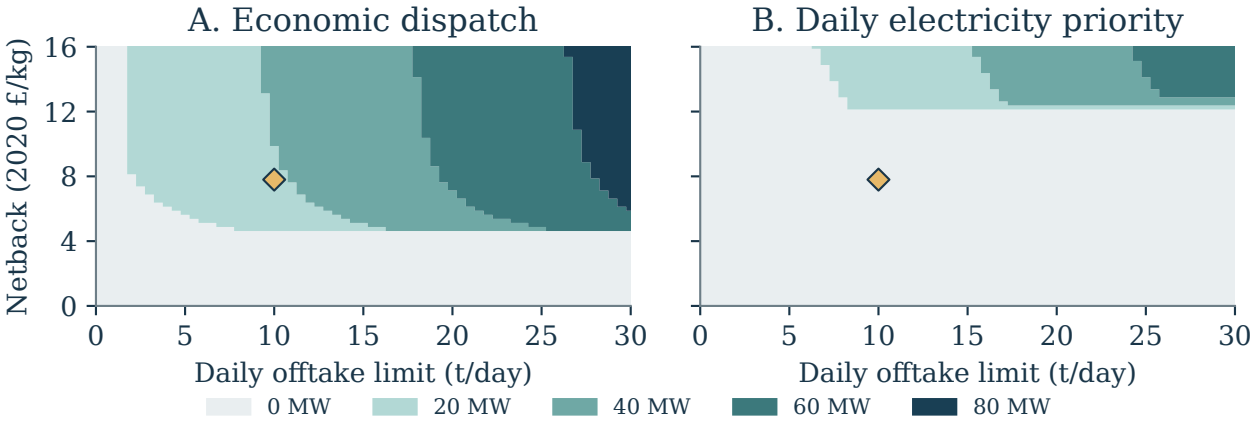


Figure 2. Optimal electrolyser capacity for alternative 2023 contracts. Diamonds mark the reference terms. The colours show installed capacity in multiples of 20 MW.

Table 3 sets annual sales floors relative to the 215.1 TWh sold in the A zero-investment case. An annual floor permits the operator to choose the days on which to preserve sales. Policy B gives electricity sales priority every day.

Table 3. Annual electricity sales retained under A. The first row is the unrestricted outcome. The remaining rows impose a minimum.

Scenario	Sales retained (%)	Sales given up (GWh)	H_2 (MW)	Gain (£m/yr)
Unrestricted	99.9266	157.92	20	8.57
Restore half the loss	99.9633	78.96	20	5.19
Tight sales floor	99.9900	21.51	20	1.04
Preserve all sales	100.0000	0.00	0	0.00

Result checks. We reconciled energy, equipment limits and revenues in 42 optimal and two rule-based schedules. Physical residuals are below 0.001 MWh. Xpress confirms eight trigger points, the capacity checks and the sales floors. A separate implementation agrees on 18 matched cases.

2.5 Robustness and returns

We hold the electrolyser at 20 MW and reoptimise dispatch on each realised calendar year. The site-only check gives background generation priority on demand and restricts the device to wind-farm output. Table 4 reports gains against the corresponding zero-investment case. Under B, the break-even netback falls to £3.13/kg in 2025 as surplus increases, but lowering both netback and offtake reverses the result. In 2023, A still gains £7.29m/year with 25% higher installation cost and an 8% real discount rate. Reoptimising on the 280 strict-quality days retains the original capacities.

Table 4. Fixed 20 MW replays. Gains are in £m/year, 2020 prices.

Period and assumptions	A. Economic dispatch	B. Daily electricity priority
2022	8.91	7.31
2023	8.57	-1.07
2024	11.95	-1.00
2025	13.17	4.48
2025, £4/kg and 5 t/day	-0.24	-0.77
2023, wind-site boundary	8.17	-1.07

Table 5. Annual value bridge for A, 2023, 20 MW.

Component	Value (£m/year)
Hydrogen netback receipts	23.97
Forgone direct electricity receipts	-12.39
Fixed operating cost	-0.52
Equivalent cost of capital and stack renewals	-2.49
Equivalent annual gain	8.57

The £29.21m upfront cost and £11.06m annual operating cash flow give a 2.64-year simple payback and a 2.90-year discounted payback. Repeating the 2023 operating cash flow for 30 years with renewals in years 11 and 22 and zero terminal value gives a net present value of £131.78m. The annual value bridge is given in Table 5.

3 Tier 1 recommendation

Keep the existing assets while verifying the buyer's netback and daily acceptance, access to the electricity needed at the site and the priority of electricity sales. Consider one 20 MW module if the contract clears the corresponding threshold. Defer battery investment at the assumed throughput and costs. Revisit it if a paid service or a different operating window can close the revenue gap.

References

[1] MMCS. *2026/27 Project Brief*, pp. 2–3 and 5–6.

[2] MMCS Group 15. *Project data repair, release 20261104-v2*. Data, methods and audit.

[3] Office for National Statistics. Consumer Prices Index, D7BT.

[4] Danish Energy Agency. Technology data for energy storage, worksheet 180 Lithium Ion Battery.

[5] International Energy Agency. *Global Hydrogen Review 2025*, electrolyser investment costs.

[6] Department for Business, Energy and Industrial Strategy. *Hydrogen Production Costs 2021* and technical assumptions annex.

[7] Department for Energy Security and Net Zero. *Hydrogen Allocation Round 1 successful projects*, 2023.